

Caden Lippie

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EDUCATION

Syracuse University, School of Information Studies — Syracuse, NY

Aug 2025 – May 2026

Master of Science, Applied Data Science, AI Concentration — GPA: 3.88

Relevant Coursework: Deep Learning, Applied Machine Learning, Cloud Management, Advanced Big Data Management

Syracuse University, David B. Falk College of Sport — Syracuse, NY

Aug 2022 – May 2025

Bachelor of Science, Sport Analytics (3+1 Accelerated Program)

Relevant Coursework: Advanced Python, Data Analysis II, R for Sport Analytics

PROFESSIONAL EXPERIENCE

Data Science Graduate Assistant — Syracuse Women's Lacrosse, Syracuse, NY

Sep 2025 – May 2026

- Envisioned and built the first-ever analytics department for the University's lacrosse program.
- Identified department objectives, recruited and interviewed prospective students, led weekly project meetings, and provided overall supervision of the first cohort of data analytics undergraduate student interns.

Data Science Intern — Syracuse Women's Lacrosse, Syracuse, NY

Jan 2023 – Aug 2025

- Designed and built a full-stack analytics platform (React, FastAPI, PostgreSQL) serving real-time performance, wellness, and testing data to coaching staff for player evaluation and injury mitigation.
- Developed a data application in R (Shiny) with custom performance metrics; presented to coaching staff and iterated on feedback to drive adoption.

Data Science Intern — Kitman Labs (Remote)

Jan 2023 – May 2023

- Analyzed multi-factor datasets (travel, injury, box-score) to quantify the effect of scheduling on team performance for a leading sports data analytics platform.
- Presented findings to senior staff and co-authored a research paper documenting results.

TECHNICAL WALKTHROUGHS

- Implemented a Transformer (encoder + decoder, per "Attention Is All You Need"), a Multilayer Perceptron, and an XGBoost model from mathematical first principles with no ML libraries — github.com/clippie

PROJECTS

Proprietary Athlete Management System

Jun 2025 – Jan 2026

- Designed a full-stack platform with a PostgreSQL database and Python (FastAPI) backend to securely manage and serve performance data.
- Built a React (JavaScript) frontend delivering responsive, user-facing dashboards for real end users.

Computer Vision Tracking System

Apr 2025 – Jun 2025

- Applied YOLOv8 transfer learning on 4,000+ manually annotated objects (players, referees, goalies) to build a real-time player-tracking system from raw game footage.

RESEARCH

- Democratizing ESG: used NLP and feature engineering on SEC 10-K filings from 300+ public companies to predict corporate sustainability scores.
- Undergraduate Thesis: modeled the impact of injuries on athlete performance using XGBoost and counterfactual simulation, built on a self-scraped 4-year dataset of ACC roster and injury records.
- Market Efficiency & Predictive Analytics: co-led research on NBA/WNBA betting strategies; presented at the Academy of Economics and Finance Conference (Feb 2024).

SKILLS

Programming: Python, R, SQL

Machine Learning & AI: PyTorch, TensorFlow, Keras, YOLO

Cloud & Data: Azure, AWS, PostgreSQL, Snowflake

Other: Git/GitHub, Excel, Tableau, Power BI